JOINT SUPERPIXEL AND SELF-REPRESENTATION LEARNING FOR SCALABLE HYPERSPECTRAL IMAGE CLUSTERING

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***Abstract*—Subspace clustering is a powerful unsupervised approach for hyperspectral image (HSI) analysis, but its high computational and memory costs limit scalability. Superpixel segmentation can improve efficiency by reducing the number of data points to process. However, existing superpixel-based methods usually perform segmentation independently of the clustering task, often producing partitions that do not align with the subsequent clustering objective. To address this, we propose a unified end-to-end framework that jointly optimizes superpixel segmentation and subspace clustering. Its core is a feedback mechanism: a self-representation network based on unfolded Alternating Direction Method of Multipliers (ADMM) provides a model-driven signal to guide a differentiable superpixel module. This joint optimization yields "clustering-aware" partitions that preserve spectral-spatial structure. Furthermore, our superpixel network learns a unique compactness parameter for each superpixel, enabling more flexible and adaptive segmentation. Experiments on benchmark HSI datasets demonstrate that our method consistently achieves superior accuracy compared with state-of-the-art clustering approaches.**

***Index Terms*—Hyperspectral image clustering, Subspace clustering, Self-representation learning, Superpixel segmentation**

I. Introduction

Hyperspectral images (HSI) capture rich spectral information across hundreds of contiguous bands, enabling fine-grained material discrimination. As a result, HSI has been widely applied in domains such as environmental monitoring and precision agriculture [1]. HSI clustering, which groups pixels into distinct categories in an unsupervised manner, is critical in HSI analysis due to the frequent lack of labeled data [2]. A variety of clustering methods have been explored, including centroid-based [3], [4], graph-based [5], hierarchical [6], and subspace-based approaches [7], [8].

Among these, subspace-based clustering has attracted increasing attention for its ability to model data as unions of low-dimensional subspaces. Representative methods include sparse subspace clustering (SSC) [7], [9], [10], low-rank representation (LRR) [8], and deep subspace clustering [11]–[13]. However, the computational and memory requirements of these methods scale quadratically with the number of pixels, severely limiting their scalability.

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To reduce this complexity, several superpixel-based subspace clustering methods have been proposed to enhance spatial consistency and efficiency [14], [15]. Yet, most existing approaches generate superpixels as a preprocessing step independent of the clustering objective [14], often leading to partitions misaligned with the downstream clustering goal and thus limiting overall performance.

While prior works like Superpixel Sampling Networks (SSN) [16] enable differentiable superpixels, they are mainly used with label supervision. In unsupervised HSI clustering, we instead use the self-representation objective as the learning signal to drive superpixel refinement, producing clustering-aware partitions aligned with intrinsic structure. As a result, our method achieves both high efficiency and strong clustering accuracy. Our main contributions are as follows:

- We propose a unified and efficient framework that jointly optimizes superpixel generation and self-representation learning, enabling clustering-oriented superpixels that align with the downstream clustering objective.
- We introduce a differentiable superpixel generation network with learnable compactness parameters, where each superpixel adaptively balances spectral and spatial characteristics for more flexible and structure-preserving segmentation.
- We design structural constraints across multiple granularities, including pixel–pixel local consistency, pixel–superpixel compactness, and superpixel–superpixel representation, which together preserve spectral–spatial structure and improve clustering performance.

II. METHODOLOGY

In this section, we present our framework that jointly optimizes superpixel segmentation and self-representation. It consists of two differentiable modules: superpixel generation and self-representation, coupled in a unified training loop. The structural feedback from the clustering module guides superpixel formation, making them not only visually coherent but also aligned with the intrinsic cluster structure. An overview is shown in Fig. 1.

A. Problem Formulation

Let $\mathbf{X} \in \mathbb{R}^{N \times D}$ denote a hyperspectral image (HSI), where N is the number of pixels and D the number of spectral bands. The objective of HSI clustering is to assign each pixel to one

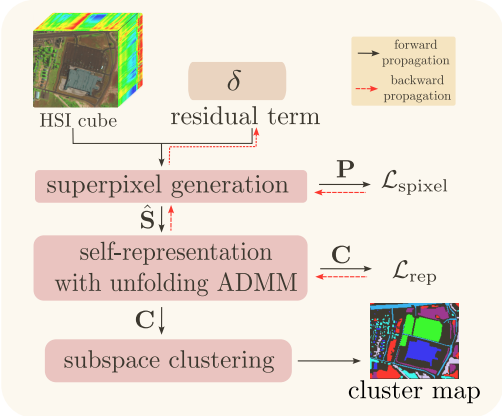


Fig. 1. Structure of the proposed method.

of $\mathcal{C}$ clusters in an unsupervised manner. Subspace-based clustering methods, which rely on the self-representation principle that each data point can be represented as a linear combination of others within the same subspace [7], have demonstrated strong performance but suffer from quadratic complexity with respect to the number of pixels. Superpixels reduce clustering units, improving efficiency while preserving spectral-spatial structure. Yet, most existing approaches generate superpixels as a preprocessing step independent of the clustering objective, often leading to partitions misaligned with the downstream clustering goal and thus limiting overall performance.

To obtain clustering-aware superpixels and improve both clustering accuracy and efficiency, we perform joint end-to-end optimization of the superpixel assignment and the sparse self-representation.

Let $\mathbf{X}' = \mathbf{X} + \boldsymbol{\delta}$ denote the adapted feature matrix, where $\boldsymbol{\delta} \in \mathbb{R}^{N\times D}$ is a learnable residual. We define $\mathbf{P} \in \mathbb{R}^{N\times M}$ as the soft assignment matrix from N pixels to M superpixels, and compute superpixel centroids $\mathbf{S} \in \mathbb{R}^{M\times D}$ as the $\mathbf{P}$-weighted means of $\mathbf{X}'$. We further obtain $\hat{\mathbf{S}}$ by column-wise ℓ_2 normalization of $\mathbf{S}^\top$, and estimate the sparse coefficient matrix $\mathbf{C} \in \mathbb{R}^{M\times M}$, where each superpixel is reconstructed by a few others, with $\operatorname{diag}(\mathbf{C}) = \mathbf{0}$ to avoid trivial solutions.

In the following subsections, we describe the formulation of the superpixel generation and self-representation modules individually, followed by the final joint optimization objective that unifies them.

B. Superpixel Generation Network

A limitation of conventional superpixel methods is that they rely on a single, fixed compactness parameter, which cannot adapt to the diverse spectral–spatial trade-offs required across different regions of an image or to the needs of the downstream clustering task. To overcome this, we introduce a learnable weight vector $\mathbf{W}$, where each element w_j controls the compactness of the j-th superpixel and is optimized jointly with the clustering objective. This design enables superpixels to adapt their balance between spectral compactness and spatial coherence in a task-driven manner, resulting in more flexible and structure-preserving partitions. The distance d_{ij} between the i-th pixel and the j-th superpixel is defined as:

$$\begin{aligned} d_{ij} = w_j \cdot \|\mathbf{X}'_i - \mathbf{S}_j\|^2 \\ + (1 - w_j) \cdot \|r(\mathbf{X}'_i) - r(\mathbf{S}_j)\|^2, \end{aligned} \tag{1}$$

where w_j is the learnable weight to balance feature similarity and spatial compactness in the j-th superpixel. The $r(\cdot)$ denotes the spatial coordinate operator for both pixels and superpixel centroids: $r(\mathbf{X}'_i)$ gives the pixel location. At the beginning of the process, the superpixel centers $\mathbf{S}_j$ and their spatial coordinates $r(\mathbf{S}_j)$ are initialized by a simple grid partitioning (or random assignment). The soft-assignment probabilities are obtained by a softmax:

$$p_{ij} = \frac{\exp(-d_{ij}/\tau)}{\sum_{n\in\mathcal{P}_i} \exp(-d_{in}/\tau)}, \tag{2}$$

where τ controls the sharpness of the distribution, and $\mathcal{P}_i$ denotes the G nearest superpixels to pixel i (fixed to G=9 by default in [16], [17]). This softmax-based assignment is fully differentiable, enabling gradients to flow through the superpixel generation process and thus supporting end-to-end training. Based on these assignments, the superpixel centers are updated by a weighted average of the assigned pixels:

$$\begin{bmatrix} \mathbf{S}_j \\ r(\mathbf{S}_j) \end{bmatrix} = \frac{1}{\sum_{i=1}^{N} p_{ij}} \sum_{i=1}^{N} p_{ij} \begin{bmatrix} \mathbf{X}'_i \\ r(\mathbf{X}'_i) \end{bmatrix}, \quad j = 1, \ldots, M. \tag{3}$$

The assignments and superpixel centers are refined iteratively, as summarized in Algorithm 1.

Algorithm 1: Iterative Superpixel Assignment

Input: Spectral feature matrix $\mathbf{X}' \in \mathbb{R}^{N\times D}$, Number of superpixels M, number of iterations T, compactness $\mathbf{W}$, temperature τ
Output: Soft assignment matrix $\mathbf{P}$, hard labels $\mathbf{L}$
Step 1: Initialization;
Initialize $[\mathbf{S}, r(\mathbf{S})]$ by grid partitioning;
Step 2: Iterative refinement;
for $t = 1$ **to** T **do**
  Compute distance matrix d_{ij} using (1);
  Obtain the soft-assignment matrix using (2);
  Update superpixel centers $[\mathbf{S}^t, r(\mathbf{S}^t)]$ using (3);
Step 3: Hard assignment;
Derive hard labels $\mathbf{L}$ from soft assignment matrix $\mathbf{P}$;
return $\mathbf{P}$, $\mathbf{L}$

To obtain high-quality superpixels, we define $\varphi(\mathbf{X}', \mathbf{P}, \mathbf{S})$ as the superpixel loss $\mathcal{L}_{\text{spixel}}$:

$$\mathcal{L}_{\text{spixel}} = \underbrace{\frac{1}{N}\sum_{i=1}^{N} \|\mathbf{X}'_i - \mathbf{F}_i\|^2}_{\text{spectral compactness}} + \underbrace{\sum_{i=1}^{N} \sum_{n\in\mathcal{N}(i)} \big(1 - \operatorname{sim}(\mathbf{P}_i, \mathbf{P}_n)\big)}_{\text{local consistency}}. \tag{4}$$

Here, $\operatorname{sim}(\cdot,\cdot)$ denotes the cosine similarity, and $\mathcal{N}(i)$ is the 4-connected neighborhood of pixel i. $\mathbf{F} \in \mathbb{R}^{N\times D}$ is obtained

by assigning each pixel to its superpixel centroid using $\mathbf{S}$ and the label map $\mathbf{L}$, so that $\mathbf{F}_i$ denotes the centroid of the superpixel containing pixel i. The first term enforces intra-superpixel compactness by pulling each pixel feature $\mathbf{X}'_i$ toward $\mathbf{F}_i$, enhancing homogeneity within superpixels. The second term promotes local label consistency. Together, these terms preserve both pixel–pixel and pixel–superpixel structure.

C. Self-representation Network (Unfolded ADMM)

After obtaining superpixels, their pairwise similarities are obtained via sparse self-representation. Following the formulation of Sparse Subspace Clustering (SSC) [7], we define $\eta(\hat{\mathbf{S}}, \mathbf{C})$ as the objective to minimize:

$$\eta(\hat{\mathbf{S}}, \mathbf{C}) = \left\|\hat{\mathbf{S}} - \hat{\mathbf{S}}\mathbf{C}\right\|_F^2 + \lambda_{\text{sr}}\|\mathbf{C}\|_1, \tag{5}$$

where the first term enforces accurate reconstruction among normalized superpixel features $\hat{\mathbf{S}}$, the second term imposes sparsity on the coefficient matrix $\mathbf{C}$. While the diagonal constraint $\text{diag}(\mathbf{C}) = 0$ is essential to prevent trivial solutions, we enforce it within the optimization process described below. In classical SSC, λ_{sr} is fixed. In unfolded formulations, however, λ_{sr} is treated as a learnable parameter, so that the sparsity level can adapt during training.

To facilitate optimization, we introduce an auxiliary variable $\mathbf{Z}$, which leads to the equivalent constrained formulation:

$$\min_{\mathbf{C},\mathbf{Z}} \|\hat{\mathbf{S}} - \hat{\mathbf{S}}\mathbf{C}\|_F^2 + \lambda_{\text{sr}}\|\mathbf{Z}\|_1, \quad \text{s.t. } \mathbf{C} = \mathbf{Z},\ \text{diag}(\mathbf{Z}) = \mathbf{0}. \tag{6}$$

To solve this constrained problem, we form the following augmented Lagrangian:

$$\mathcal{L}(\mathbf{C}, \mathbf{Z}, \boldsymbol{\mu}) = \|\hat{\mathbf{S}} - \hat{\mathbf{S}}\mathbf{C}\|_F^2 + \lambda_{\text{sr}}\|\mathbf{Z}\|_1 + \langle \boldsymbol{\mu}, \mathbf{C} - \mathbf{Z}\rangle + \frac{\rho}{2}\|\mathbf{C} - \mathbf{Z}\|_F^2, \tag{7}$$

where $\boldsymbol{\mu}$ is the dual variable and $\rho > 0$ is the penalty parameter. The optimization proceeds with iterative updates:

C-update:

$$\mathbf{C}^{(k+1)} = (2\hat{\mathbf{S}}^T\hat{\mathbf{S}} + \rho\mathbf{I})^{-1}\Big(2\hat{\mathbf{S}}^T\hat{\mathbf{S}} - (\boldsymbol{\mu}^k - \rho\mathbf{Z}^k)\Big). \tag{8}$$

Z-update:

$$\mathbf{Z}^{(k+1)} = \text{ReLU}(|\mathbf{C}^{(k+1)} + \frac{\boldsymbol{\mu}^k}{\rho}| - \frac{\lambda_{\text{sr}}^k}{\rho}) \cdot \text{sgn}(\mathbf{C}^{(k+1)} + \frac{\boldsymbol{\mu}^k}{\rho}), \tag{9}$$

$$\mathbf{Z}^{(k+1)} = \mathbf{Z}^{(k+1)} - \text{diag}(\mathbf{Z}^{(k+1)}), \tag{10}$$

where $\text{ReLU}(\cdot) = \max(0, \cdot)$ is the standard rectified linear unit for the element-wise soft-thresholding.

μ-update:

$$\boldsymbol{\mu}^{(k+1)} = \boldsymbol{\mu}^k + \rho\big(\mathbf{C}^{(k+1)} - \mathbf{Z}^{(k+1)}\big). \tag{11}$$

Through the differentiable ReLU-based thresholding and the introduction of learnable λ_{sr}, the ADMM updates are unfolded into a finite-depth neural network, where each iteration corresponds to one layer. This formulation embeds the optimization process into the network architecture and enables end-to-end training. According to the self-representation objective defined in Eq. (5), we impose the following losses.

Reconstruction loss.

$$\mathcal{L}_{\text{recon}} = \left\|\hat{\mathbf{S}}\mathbf{C} - \hat{\mathbf{S}}\right\|_F^2, \tag{12}$$

which enforces accurate reconstruction of superpixels using the self-representation matrix.

Sparsity regularization. We further constrain the coefficient matrix $\mathbf{C}$ with two complementary terms:

$$\mathcal{L}_{\ell_1} = \|\mathbf{C}\|_1, \quad \mathcal{L}_{\text{entropy}} = -\frac{1}{M}\sum_{j=1}^{M}\sum_{i=1}^{M}\tilde{c}_{ij}\log(\tilde{c}_{ij} + \varepsilon), \tag{13}$$

where $\tilde{c}_{ij}$ is the normalized absolute value of c_{ij}, and ε is a stability constant, empirically set to 10^{-8}. The ℓ_1 term suppresses weak connections, and the entropy term promotes peaked distributions and improves clustering robustness. The final objective of the self-representation module for uncovering superpixel-level structure is formulated as a weighted sum of $\mathcal{L}_{\text{recon}}$, $\mathcal{L}_{\ell_1}$, $\mathcal{L}_{\text{entropy}}$. For simplicity, we set:

$$\mathcal{L}_{\text{rep}} = 2 \cdot \mathcal{L}_{\text{recon}} + \mathcal{L}_{\ell_1} + \mathcal{L}_{\text{entropy}}, \tag{14}$$

and refer to it as the self-representation loss.

D. Joint Optimization Objective

Having defined the individual components and their respective objectives, we now present the overall joint optimization problem. Our framework minimizes a unified objective that integrates the superpixel generation φ, the self-representation loss η, and the noise regularization term:

$$\min_{\mathbf{P},\mathbf{S},\mathbf{C}} \mathcal{L}_{\text{spixel}} + \alpha\mathcal{L}_{\text{rep}} + \mathcal{L}_{\text{noise}}, \quad \text{s.t. } \text{diag}(\mathbf{C}) = \mathbf{0}. \tag{15}$$

Here, $\mathcal{L}_{\text{noise}}$ regularizes the learnable residual term $\boldsymbol{\delta}$ added to the input features to prevent overfitting. We define it as:

$$\mathcal{L}_{\text{noise}} = \frac{\lambda_{\text{noise}}}{ND}\|\boldsymbol{\delta}\|_F^2, \tag{16}$$

where ND is the size of $\boldsymbol{\delta}$ and λ_{noise} is a parameter. In all experiments, we fix $\lambda_{\text{noise}} = 50$. The only tunable parameter is α, whose effect will be examined in the next section, as its value depends on the dataset.

III. Experiments

To demonstrate the effectiveness of the proposed method, we evaluate its performance through experiments on widely used benchmark hyperspectral datasets. All implementation details are provided in our source code[1].

A. Experiment Setup

In this section, we evaluate the proposed method on three benchmark datasets described in Table I.

The number of superpixels is set as the maximum of two values: (1) the edge-based estimation in [14], which reflects scene complexity, and (2) a lower bound $50\mathcal{C}/\mathcal{R}$, where $\mathcal{C}$ is the number of classes and $\mathcal{R}$ is the area ratio of the cluster region to the entire image. The factor 50 is an empirical choice to ensure sufficient samples per class [7]. This dual criterion makes the number of superpixels both adaptive to spatial complexity and robust across different class distributions.

[1]https://github.com/lxlscut/superpixel_ssc

TABLE I
SUMMARY OF THE HSI DATASETS USED IN THE EXPERIMENTS.

Dataset	Bands	Size (pixels)	Classes	Labeled Samples
Trento	63	600×166	6	30,214
Salinas	204	512×217	16	54,129
Urban	162	150×160	7	12,048

B. Results

1) Comparison with other methods: In our experiments, we compare the proposed method with several widely adopted clustering approaches, including the centroid-based methods FCM [4], DEKM [18], the graph-based methods SpectralNet [19], the hierarchical method FINCH [6], the deep clustering method IDEC [20], and the basis-representation-based subspace clustering methods MADL [21]. In addition, we include a comparison with the superpixel-based subspace clustering method NCSC [14]. Specifically, the source codes for IDEC [20] and DEKM [18] are obtained from the ClustPy library [22], while the implementations of the other recent methods are taken from their official repositories. The results of the experiments are shown in Table II.

TABLE II
CLUSTERING PERFORMANCE COMPARISON RESULTS.

Method	Salinas			Trento			Urban		
	OA (%)	NMI	κ	OA (%)	NMI	κ	OA (%)	NMI	κ
K-means	71.43	0.7829	0.6840	83.54	0.7439	0.7829	72.59	0.6932	0.6733
FCM	65.57	0.7597	0.6257	75.63	0.7191	0.6909	78.06	0.7065	0.7397
FINCH	65.63	0.7941	0.6099	81.37	0.7932	0.7457	72.23	0.7060	0.6728
DEKM	74.92	**0.8635**	0.7176	85.48	0.8047	0.8018	74.30	0.7454	0.6933
SpectralNet	69.87	0.7904	0.6707	68.35	0.7764	0.5982	69.13	0.7183	0.6326
IDEC	53.48	0.7188	0.4949	69.69	0.7155	0.6246	70.22	0.6804	0.6503
MADL	51.95	0.6928	0.4560	88.72	**0.9292**	0.8490	72.49	0.7166	0.6740
NCSC	55.29	0.6409	0.5066	66.17	0.5160	0.5548	53.61	0.4742	0.4481
Ours	**80.96**	0.8354	**0.7868**	**90.21**	0.8297	**0.8694**	**84.52**	**0.7517**	**0.8173**

The baseline methods show varying strengths: DEKM performs well on Salinas, while MADL achieves strong results on Trento but fails on Salinas. The superpixel-based NCSC lags behind across all datasets. In contrast, our method consistently attains the highest OA and κ, while also ranking among the top in NMI, demonstrating clear advantages in accuracy, robustness, and generalization.

In contrast to pixel-level subspace clustering ($\mathcal{O}(N^2)$), which suffers from out-of-memory (OOM) errors on large datasets, deep clustering methods achieve linear complexity ($\mathcal{O}(N)$) and low memory usage via mini-batch training (e.g., on Salinas: SpectralNet 380MB/134s, IDEC 250MB/307s). Both NCSC and our method leverage superpixels ($\mathcal{O}(M^2)$) for scalability. Although our joint optimization (3900MB/240s) incurs a marginal overhead compared to the static NCSC (2530MB/223s), it remains fully scalable on standard GPUs, justifying the cost with significantly superior clustering performance.

2) Ablation Study: To assess the contribution of the self-representation module, we conduct an ablation study with five model variants: M1, direct clustering on initial superpixels without training; M2, training only the superpixel generation network with $\mathcal{L}_{\text{spixel}}$ and $\mathcal{L}_{\text{noise}}$; M3, training only the unfolded ADMM network with $\mathcal{L}_{\text{rep}}$; M4, training superpixel generation network and unfolded ADMM network separately; and Ours, the full model jointly optimized with all three losses. The corresponding results are presented in Table III.

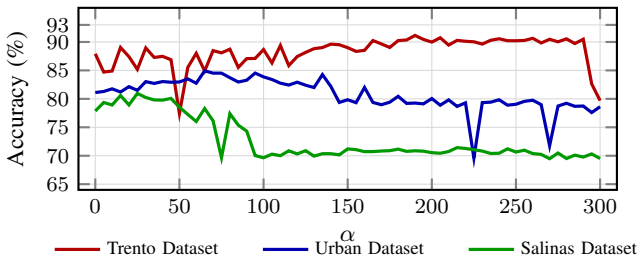


Fig. 2. Impact of α on clustering accuracy.

TABLE III
ABLATION STUDY RESULTS.

Model Variant	Salinas			Trento			Urban		
	OA (%)	NMI	κ	OA (%)	NMI	κ	OA (%)	NMI	κ
M1	78.90	0.8220	0.7639	85.09	0.7884	0.8011	80.95	0.7239	0.7746
M2	77.86	0.8251	0.7523	87.92	0.8045	0.8384	81.11	0.7243	0.7765
M3	79.60	0.8332	0.7716	85.41	0.8119	0.8054	80.73	0.7234	0.7719
M4	77.51	0.8212	0.7483	87.94	0.7972	0.8387	80.82	0.7203	0.7729
Full model (Ours)	**80.96**	**0.8354**	**0.7868**	**90.21**	**0.8297**	**0.8694**	**84.52**	**0.7517**	**0.8173**

The ablation results show that only the joint optimization in our full model gets the best performance. This indicates that the joint optimization yields clustering-oriented superpixels, which in turn increase clustering performance across datasets.

C. Parameter Study

We evaluate the impact of the self-representation loss weight α on clustering performance, as illustrated in Fig. 2. Clearly, the proposed method is highly robust with respect to this parameter on Trento, where accuracy stabilizes above 89% once $\alpha > 180$ and only declines at very large α values. On Urban, accuracy at first improves gradually with increasing α, peaks when α is between 65 and 100, and then decreases. By contrast, Salinas is more sensitive, reaching optimal performance for α in the range [25,45]. This is attributed to its high spectral dimensionality (204 bands), which increases the reconstruction magnitude; if α is too large, this term dominates gradients, weakening spatial compactness. These results suggest that simpler datasets can tolerate larger α values, while more complex ones require smaller values.

IV. CONCLUSION

In this work, we introduced a unified framework that jointly optimizes superpixel generation and self-representation for hyperspectral image clustering. By coupling a differentiable superpixel module with an unfolded ADMM network through structural feedback, the model produces clustering-aware partitions that improve representation quality. The framework further integrates multi-granularity constraints for spectral–spatial structure preservation. Experiments on multiple benchmarks confirm consistent improvements over existing methods, demonstrating the effectiveness and robustness of the proposed approach.

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